

Computer Science & IT

Theory of Computation



Context Free Languages

Lecture No. 4



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Recap of Previous Lecture



Topic

CFG

Topic

Topic

Topics to be Covered



Topic

CFG

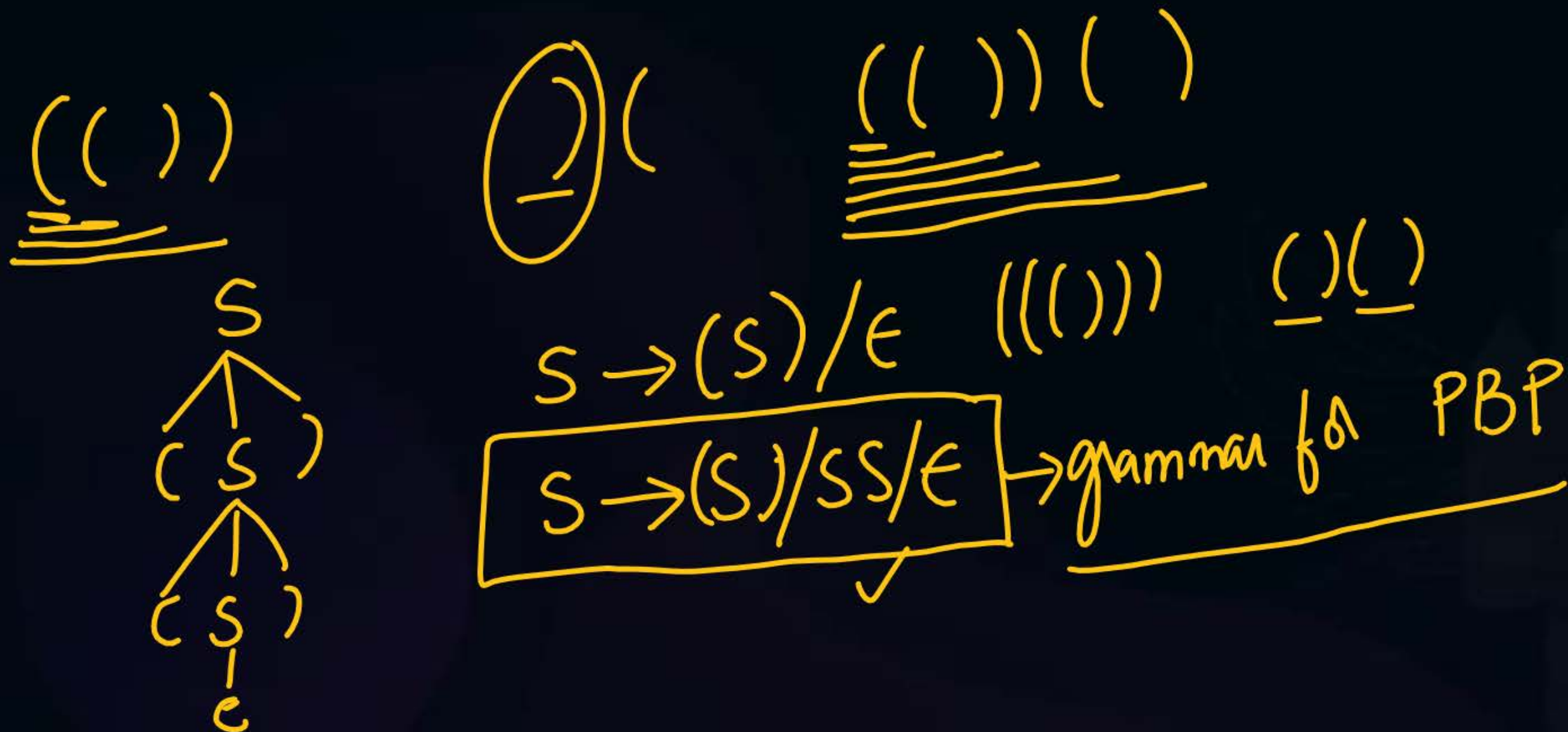
Topic

Topic

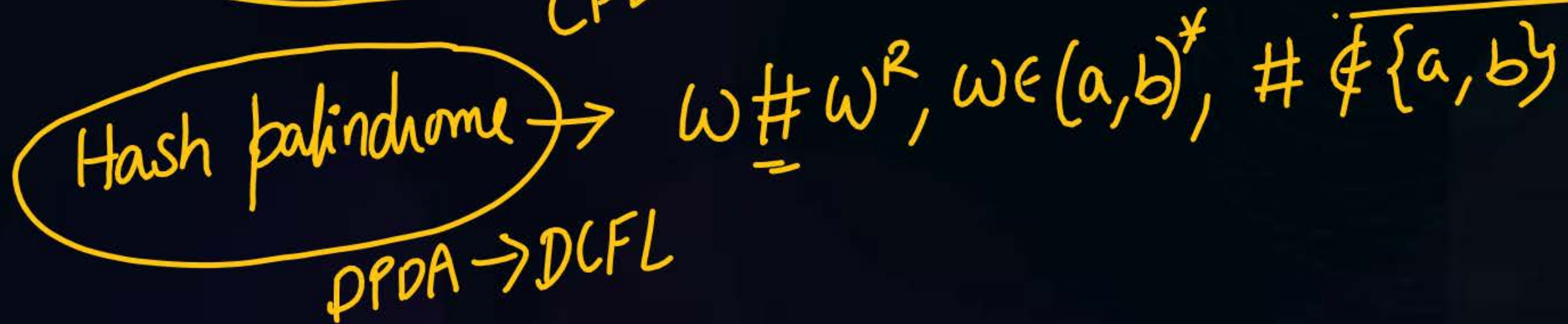
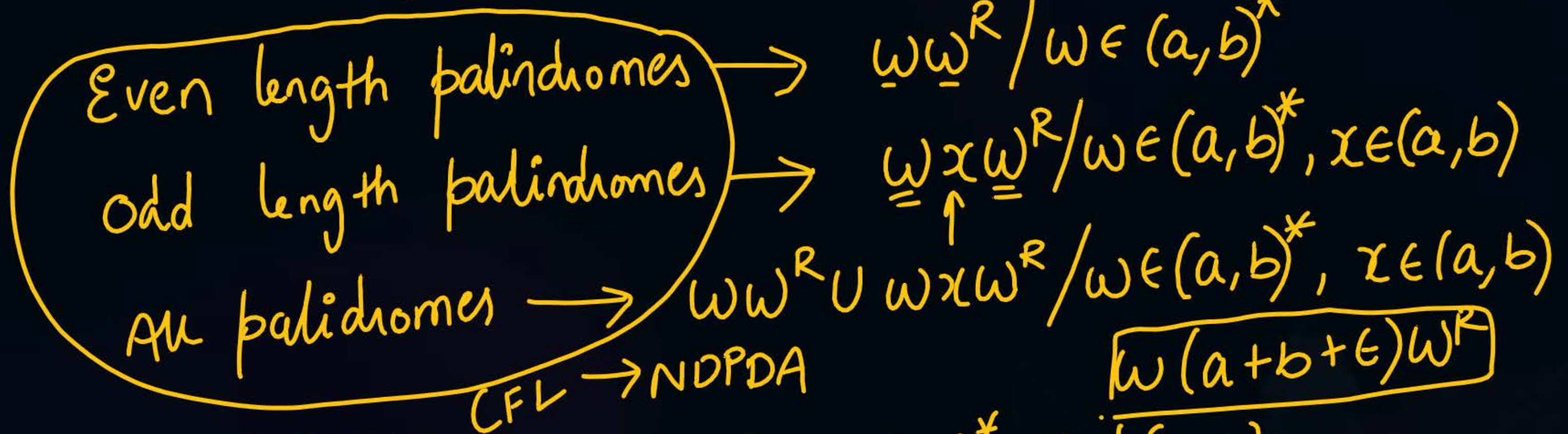
Perfectly balanced parenthesis:-



$= w / n_c(w) = n_o(w)$ if v is any prefix of w , $n_c(v) \geq n_o(v)$



string matching (Palindromes)



Even pal: $a \overbrace{b b b}^{} a$

$S \rightarrow aSa / bSb / \epsilon$

odd pal: $S \rightarrow aSa / bSb / \underline{a} / \underline{b}$

Any pal: $S \rightarrow aSa / bSb / a / b / \epsilon$

Hash pal: $w \# w^R$
 $S \rightarrow aSa / bSb / \underline{\#}$

2 min break
Throat problem

$$ww^R \cup \underline{(0,1)^*} / w \in (0,1)^*$$

$$\Downarrow$$

$$(0,1)^* \rightarrow \text{Regular}$$

$$0^n 1^n \cup (0,1)^*$$

$$\Downarrow$$

$$(0,1)^* \rightarrow \text{Regular}$$

$\frac{ww^R}{w \in (0,1)^*} \rightarrow XReg, XDGL, VCFL$
 $\downarrow$
 $NPDA \rightarrow CFL$

$$ww^R / w \in a^*$$

$$\left\{ \epsilon, \begin{array}{cc} aa \\ \downarrow \quad \downarrow \\ w \quad w^R \end{array}, \begin{array}{cc} aa & aa \\ \downarrow & \downarrow \\ w & w^R \end{array} \dots \right\}$$

$$(aa)^* \checkmark$$

$$\underline{\underline{\omega(\omega^R)^*}} / \underline{\underline{\omega \in a^*}}$$

$$\underline{\underline{\omega}} \cdot \{ \epsilon, \omega^R, (\omega^R)^2, (\omega^R)^3, \dots \}$$

$$\omega, \omega\omega^R, \omega(\omega^R)^2, \omega(\omega^R)^3, \dots$$

$$\begin{array}{c} \omega \\ \downarrow \\ \textcircled{a^*} \end{array} \rightarrow \underline{\underline{\text{Reg}}}$$

$$w \# w^R / w \in \underline{a}^*$$

#, a#a, aa#aa, aaa#aaa

$$\checkmark \underline{a}^n \# \underline{a}^n / \underline{n \geq 0}$$

DPDA
 $\checkmark$ (DCFL)
 $\checkmark$ (CFL)



$ww^R / w \in \{a, b\}^*, |w| \leq 10$

finite $\rightarrow$ Regular

$w.w.w / w \in \underline{(a, b)}$

$w=a$ $w=b$

$aaa + bbb$

$$\omega(\omega^R)^* / \omega \in (a,b)^*$$

$$\omega \cdot \{ \epsilon, \omega^R, (\omega^R)^2, (\omega^R)^3, \dots \}$$

$$\omega, \omega\omega^R, \omega(\omega^R)^2, \dots$$

$$\downarrow$$

$$\underline{(a+b)^*} \rightarrow \text{Reg}$$

$$w^* w^R / w \in (a, b)$$

$$w = \underline{a}, w = b$$

$$a^* a^R + b^* b^R$$

$$a^* a + b^* b$$

$$a^+ + b^+$$

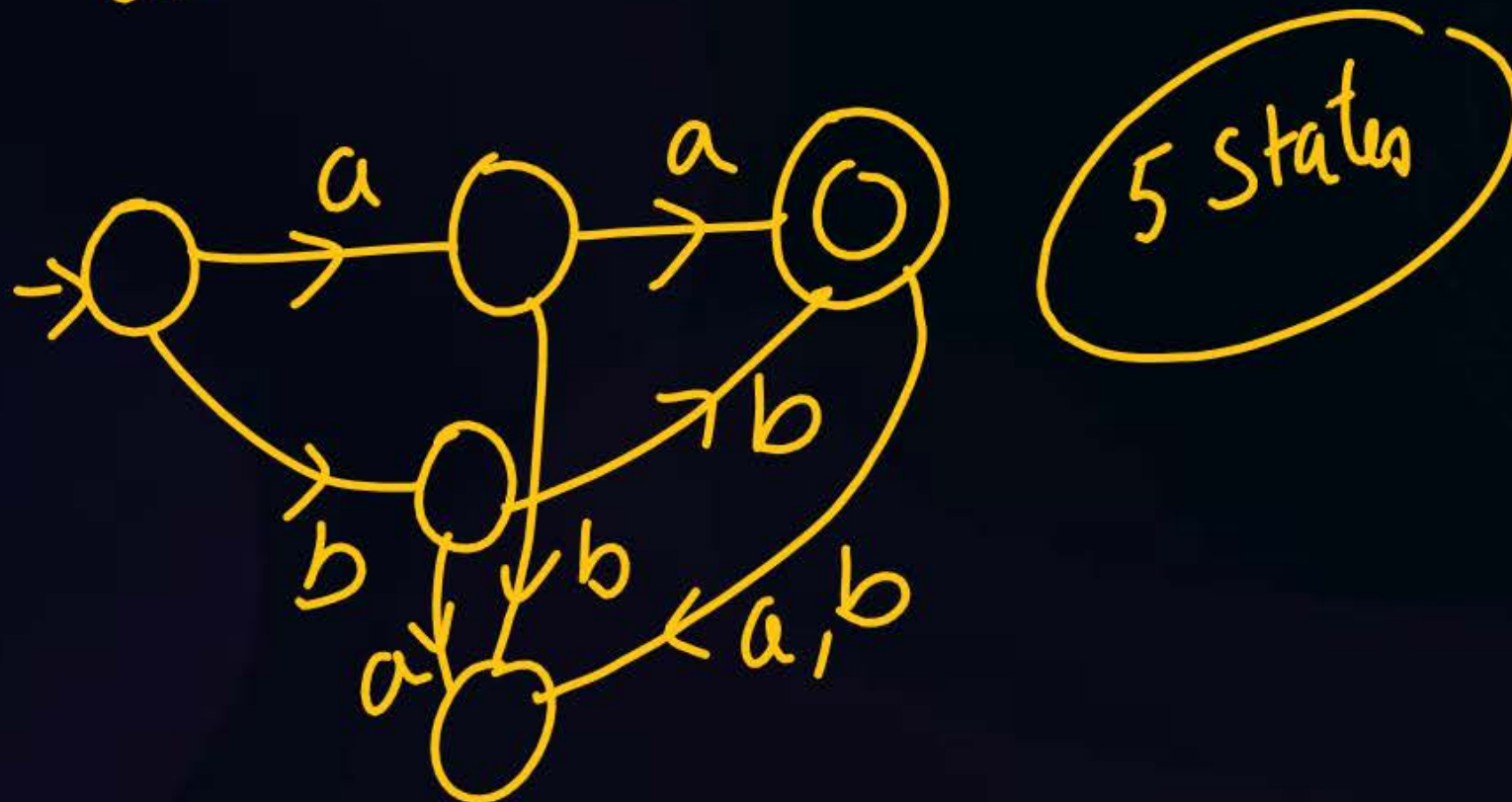
$$\begin{array}{c}
 \begin{array}{c} \circ \\ \otimes \end{array} w^R / \underline{w} \in \underline{(a,b)^*}^* \\
 \downarrow \quad \downarrow \\
 \epsilon \quad w^R \\
 \downarrow \\
 (a+b)^* \rightarrow \text{Reg}
 \end{array}$$

$ww^R / w \in \{a, b\} \rightarrow$ How many states are there in min DFA?

$w=a \rightarrow aa$

$w=b \rightarrow bb$

$aa + bb$



$w(w^R)^* / w \in (a,b)^+ \rightarrow$ doesn't contain 'e'

$w \cdot \{ \epsilon, w^R, (w^R)^2, (w^R)^3, \dots \}$

$w, ww^R, w(w^R)^2, w(w^R)^3, \dots$



$$\begin{array}{c}
 wxw^R / w \in (a,b)^* \quad x \in (a,b)^* \\
 \downarrow \quad \downarrow \\
 \epsilon \quad \epsilon \\
 \downarrow \\
 (a+b)^* \rightarrow \underline{\text{Regular}}
 \end{array}$$



$$w \check{x} w^R / w \in (a,b)^*, \underline{\underline{x \in (a,b)^+}}$$

$$\begin{array}{c} | \\ \epsilon \end{array} \quad \begin{array}{c} | \\ \epsilon \end{array}$$

$$(a+b)^+ \checkmark$$

✓

Lang doesn't contain ϵ



$$\underline{wxw^R} / \underline{w \in (a,b)^*} \quad \underline{x \in (a,b)}$$

odd length palindromes

NDPDA $\rightarrow$ CFL
xReg

$$\begin{array}{ccc} \checkmark & & \checkmark^R \\ w & x & w^R \\ | & | & | \\ \epsilon & \downarrow & \epsilon \\ & (a+b) & \\ & \underline{(a+b)} & x \end{array}$$

$$w x w^R / w \in (\underline{a}, \underline{b}) \quad \underline{x} \in \underline{(a, b)^*}$$

$$w = \underline{a} \quad w = b$$

$$a \underline{x} a \quad b x b$$

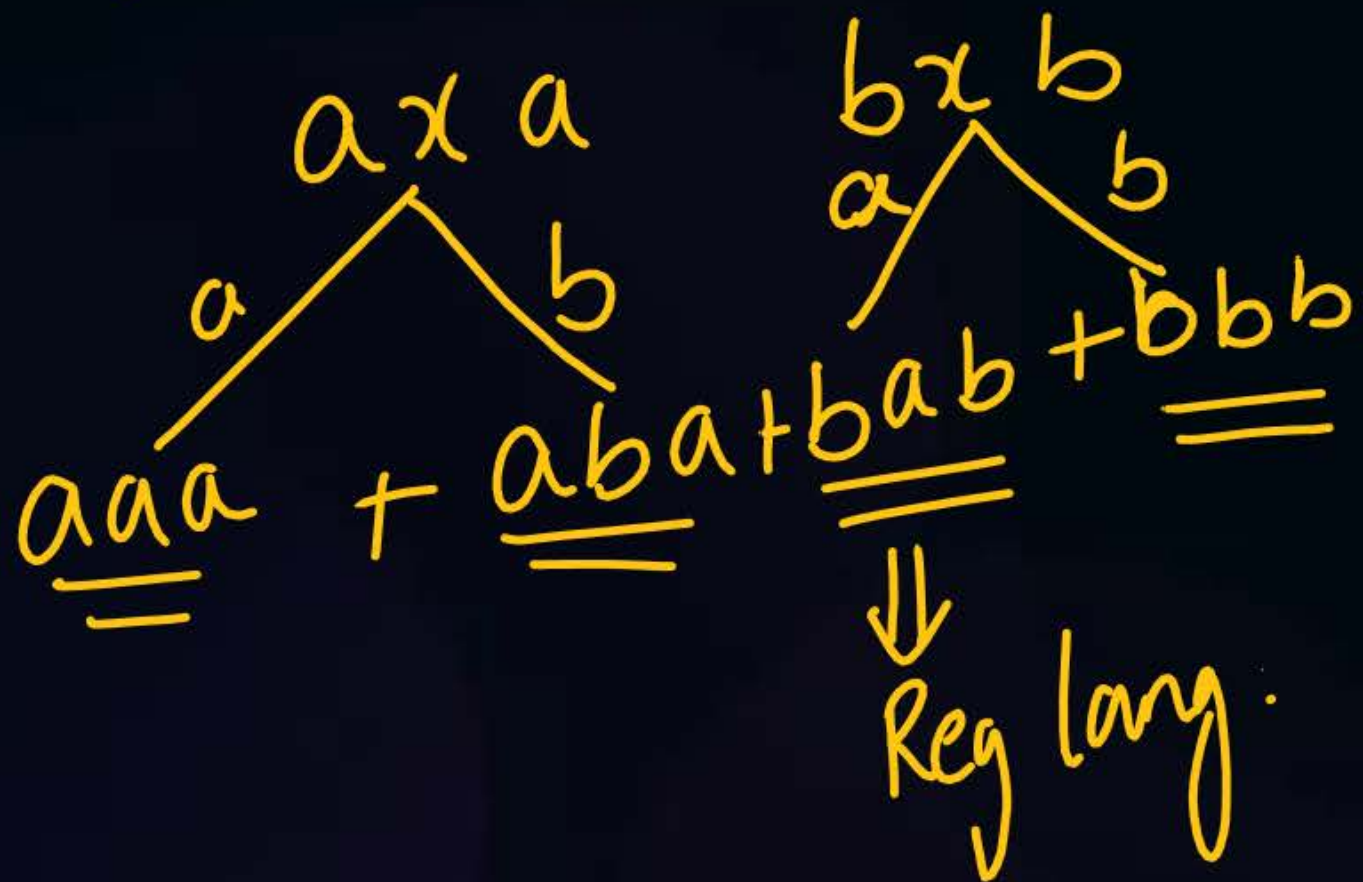
$$\underline{a} (a+b)^* \underline{a} + \underline{b} (a+b)^* \underline{b}$$

↓
Reg

$$wxw^R / w \in (a,b) \quad x \in (a,b)$$

$$w = a$$

$$w = b$$



$$L = wxw^R / w \in (a,b)^+ \quad x \in (a,b)^*$$

$$L_1 = \underbrace{a(a+b)^*a}_{w \cdot x \cdot w^R} \quad L_2 = \underbrace{b(a+b)^*b}_{w \cdot x \cdot w^R}$$

$$L \stackrel{?}{=} L_1 + L_2$$

$$w = \underline{ab}$$

$$a \underbrace{b(a+b)^*b}_{w \cdot x \cdot w^R} a$$

$$w = ba$$

$$\underline{b} \underbrace{a(a+b)^*a}_{w \cdot x \cdot w^R} b$$

$$a \underbrace{b(a+b)^*bb}_{w \cdot x \cdot w^R} a$$

$$\underline{\underline{a(a+b)^*a}}$$

$$L = \underline{xww^R} / x \in (a,b)^+ w \in (a,b)^+$$

↓
NPDPA ✓

↓
(CFL) ✓ xww^R

$$L_1 = ((a+b)^+ a a) \checkmark$$

$$L_2 = ((a+b)^+ b b) \checkmark$$

xReg ✓ CFL

$$\underline{L} \stackrel{?}{=} \underline{L_1 + L_2}$$

$$w = ab \quad ((a+b)^+ ab)ba \Rightarrow ((a+b)^+ \underline{ba}) \underline{w w^R} \checkmark$$

$$L = wxw^R / x, w \in (a+b)^+$$

$$L_1 = \frac{w \quad x \quad w^R}{\quad | \quad | \quad \backslash}$$

$$L_1 = \frac{a \quad (a+b)^+ \quad a}{\quad | \quad | \quad \backslash}$$
 ✓✓

$$L_2 = \frac{w \quad x \quad w^R}{\quad | \quad | \quad \backslash}$$

$$L_2 = \frac{b \quad (a+b)^+ \quad b}{\quad | \quad | \quad \backslash}$$
 ✓

$$L \stackrel{?}{=} L_1 \cup L_2$$

$$w = ab$$

$$a \quad \underbrace{b \quad (a+b)^+ \quad b}_{\quad | \quad | \quad \backslash} \quad a$$

$$\underline{a \quad (a+b)^+ \quad a}$$

$$w = ba$$

$$b \quad \underbrace{a \quad (a+b)^+ \quad a}_{\quad | \quad | \quad \backslash} \quad b$$

$$b \quad (a+b)^+ \quad b$$

Derivation:-

The process of deriving a string from the grammar is called derivation.

Left most derivation:-

In every step of derivation, if there is a choice in variable substitution and left most variable is substituted, then it is LMD.

✓ LMD. $S \rightarrow \underline{A}BC \rightarrow a\underline{B}C \rightarrow ab\underline{C} \rightarrow abc$

✓ RMD: $S \rightarrow AB\underline{C} \rightarrow A\underline{B}C \rightarrow \underline{A}bC \rightarrow \underline{a}bc \checkmark$

→ For every derivation tree, there is only 1 LMD and 1 RMD possible.

→ If $w \in L$, there will be at least one derivation tree, at least 1 RMD and at least 1 LMD possible



→ A grammar is unambiguous if for every string in the language, there is only one derivation tree, only 1 LMD and only 1 RMD.

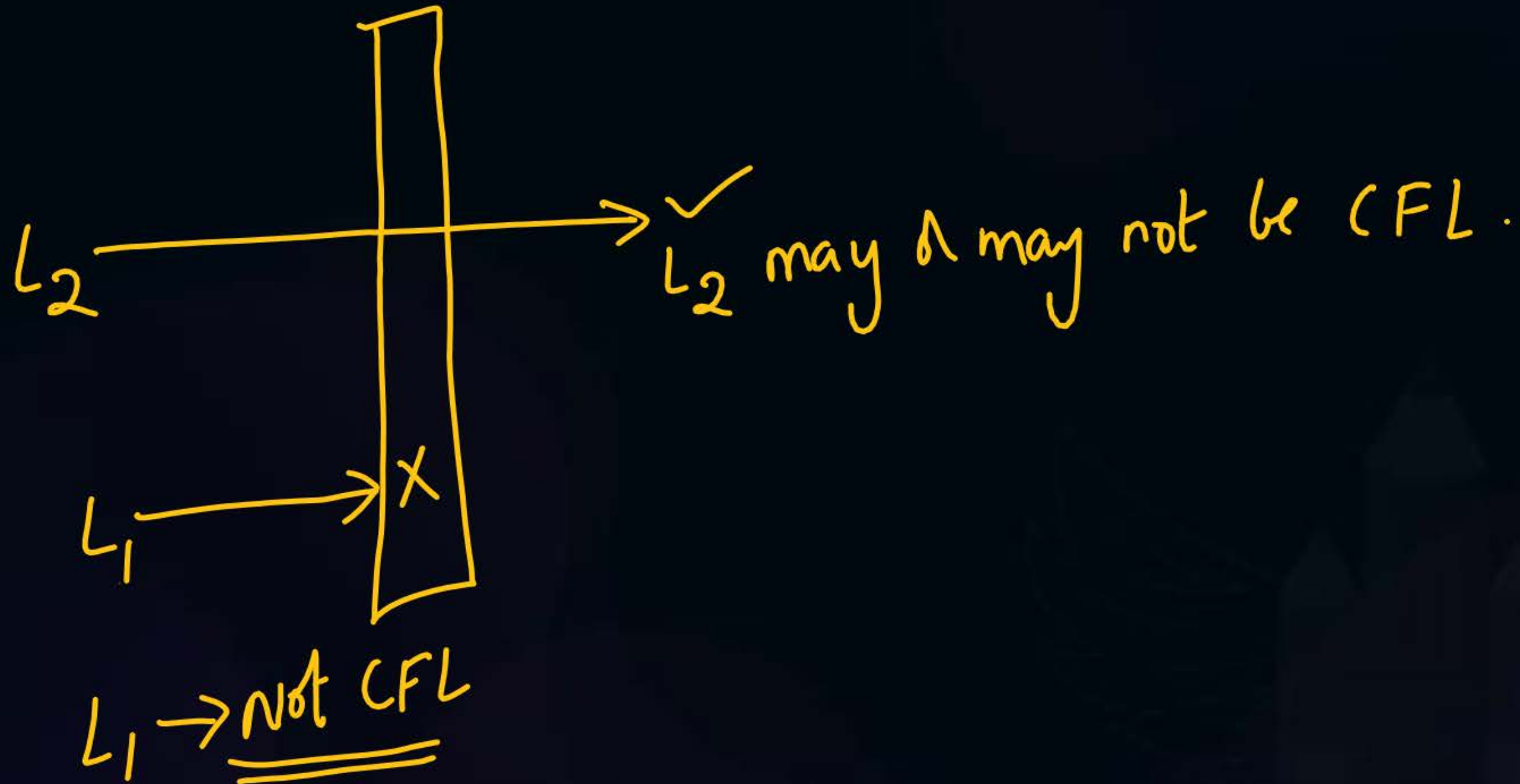


Ambiguous grammar:-

A grammar G is said to be ambiguous grammar
 if $\exists w \in L$, such that w has more than one parse tree.
 So there will be more than one LMO, more than one
 RMD for w .

In unambiguous grammars, every string has
 exactly one derivation tree, one LMD, one RMD
 $\rightarrow$ If w has one derivation tree $\Leftrightarrow w$ has 1 RMD
 $\Leftrightarrow w$ has 1 LMD

Pumping lemma for CFG



Pumping lemma for CFL:-



Let L be a CFL and $w \in L$, such that
 $|w| \geq n$ for some positive integer n , then
' w ' can be decomposed as $w = abcde$

Such that

- i) $bd \neq \epsilon$
- ii) $|bcd| \leq n$
- iii) For all $i \geq 0$ string $ab^i c d^i e$ is also in language

Prove that $a^n b^n c^n / n \geq 1$ is not CFL



$a a a | \underline{a a} | \underline{b b b b b} | \underline{c c} | c c c$

$a a a | \underline{a b} | b b | \underline{b c} | c c c$

$a b a b b c b c \checkmark$

Not CFL ✓
/



THANK - YOU